

Akshay Aradhya

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CAREER OBJECTIVE

Motivated Data Science enthusiast pursuing BE in Electronics and Communication Engineering, seeking a Data Science / ML Internship to apply analytical and machine learning skills.

EDUCATION

Bachelor of Engineering (B.E),

Sri Jayachamarajendra College of Engineering Mysore

2022 - 2026

TRAININGS / CERTIFICATIONS

Machine Learning And Data Science Training

Jul 2025 - Jan 2026

Online / Self Learning, Virtual

Skills Learned:

- Python for Data Science
- NumPy and Pandas
- Data Visualization using Matplotlib and Seaborn
- Machine Learning algorithms (Logistic Regression, Decision Tree, Random Forest)

PORTFOLIO

[GitHub link ↗](#)

PROJECTS

Credit Card Fraud Detection using Machine Learning

Mar 2026

Developed a machine learning classification model to detect fraudulent credit card transactions by analyzing transaction patterns and identifying anomalies. The dataset contained highly imbalanced data, where fraudulent transactions were very rare compared to normal transactions.

To address this challenge, preprocessing techniques were applied to handle the class imbalance problem and improve model performance. Various machine learning algorithms were used to train the model, and the performance was evaluated using precision, recall, and F1-score, which are important metrics for fraud detection.

The goal of the project was to accurately detect fraudulent transactions while minimizing false positives, helping financial institutions reduce financial losses and improve transaction security.

Technologies Used:

Python, Pandas, NumPy, Scikit-learn, Machine Learning Algorithms

IoT-Based Solar-Powered Multipurpose Agri-Bot

Aug 2025 - Feb 2026

Developed an IoT-based solar-powered agricultural robot using the ESP32 microcontroller to assist farmers in performing multiple field operations efficiently. The bot was designed to perform tasks such as automatic irrigation, seed sowing, and real-time field monitoring.

Solar energy was used as the primary power source to make the system energy-efficient and environmentally sustainable.

Various IoT sensors were integrated to monitor environmental conditions such as soil moisture, temperature, and humidity. Based on the sensor data, the system can automatically trigger irrigation when the soil moisture level falls below a certain threshold.

The ESP32 module enabled wireless communication, allowing sensor data to be monitored remotely through an IoT platform. This project helps reduce manual labor and improves precision farming by providing smart automation in agriculture.

• Technologies Used:

ESP32, IoT Sensors, Solar Panel, Embedded Systems, Wireless Communication

Breast Cancer Detection using Machine Learning

Feb 2026

Developed a machine learning classification model to predict whether a tumor is malignant (cancerous) or benign (non-cancerous) using medical diagnostic data.

The project involved performing data preprocessing, feature selection, and normalization to improve the quality of the dataset and enhance model accuracy. Important features related to tumor characteristics were selected to build an efficient predictive model.

Multiple machine learning algorithms were trained and evaluated to determine the best-performing model. The system helps assist medical professionals by providing early prediction and diagnosis support for breast cancer detection.

Technologies Used:

Python, Pandas, NumPy, Scikit-learn, Machine Learning Algorithms

[Sonar Signal Classification using Machine Learning](#)

[↗](#)

Jan 2026

Developed a machine learning classification system to identify whether an underwater object detected by sonar signals is a Rock or a Mine. The project uses sonar signal frequency measurements obtained from sonar returns reflected by objects underwater.

A Support Vector Machine (SVM) classifier was implemented to learn patterns from the sonar signal data and accurately classify objects. The model was trained on labeled data and evaluated using standard classification metrics to measure its prediction performance.

This project demonstrates how machine learning can be applied in underwater object detection and defense applications, helping distinguish between harmless rocks and dangerous mines.

Technologies Used:

Python, NumPy, Pandas, Scikit-learn, Support Vector Machine (SVM)

SKILLS

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|-----------------------|--------------------|-----------------|
| • Python | • SQL | • C# |
| • Pandas | • NumPy | • scikit-learn |
| • Matplotlib | • Seaborn | • VS Code |
| • Google Colab | • Power BI | • MySQL |
| • Statistics | • Data Cleaning | • Data Analysis |
| • Feature Engineering | • Model Evaluation | |

EXTRA CURRICULAR ACTIVITIES

- Technical Activities
 - Participated in machine learning workshops and coding practice.
 - Practiced Python programming and problem solving.

ADDITIONAL DETAILS

- Programming: Python Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn Machine Learning: Classification, Model Evaluation, Feature Selection Tools: VS Code, Google Colab, MATLAB IoT: ESP32 Microcontroller, Sensors, Embedded Systems